Shanmugam Subramanian

Mechanical Engineer II | Precision Electromechanical Design | Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience in precision electromechanical design for antenna and radar hardware. Strong in GD&T, tolerance stack-up, precision fixtures and jigs, and vibration and thermal analysis, with a 90% part-wide first-article inspection pass rate across CNC, EDM, sheet-metal and molded parts. Targeting mechanical design roles in semiconductor capital equipment.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

- Released CNC, EDM, sheet metal, die-cast, injection-molded, RF PCB, & FDM parts with a **90% FAI pass rate**.
- Developed a **pneumatic-actuated pivot punching fixture** that automated a 2-person task to one operator and **improved yield from 60% to 95%** (saving $0.4 per failed cycle).
- Designed **precision assembly fixtures**, rivet press-pull tools, & alignment jigs for RF PCB sub-assemblies: **40% rework reduction, 75% faster riveting, and up to 20–50 mins saved per antenna**.
- Performed **harmonic & random vibration analyses** with 80% first-pass correlation to physical tests (IEC 60068-2-6/64).
- Executed **static structural analysis** with FOS targets of 1.5–3 (metals) & 2–5 (plastics).
- Built Excel-based **hand-calculation tools** to estimate drag force from geometry, crimping force for pneumatic cylinder sizing, and snap-joint behavior in plastic part design.
- Performed **IEC 60068-2-31 drop tests** for mechanical qualification of antenna products.
- Owned end-to-end **NPD & NPI** for eight 5G Macro & Distributed Antenna Systems products for the US & Canada markets.
- Authored a **utility patent draft** for an injection-molded, snap-fit PCB/sheet-metal support pillar (twist-lock base, flexing offset arms, two-cavity mold).
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.
- Managed **ECRs, FAIs, & SVPRs** using Salesforce Propel **PLM** & SolidWorks **PDM** in coordination with the global team.
- Introduced topology-optimized FDM 3D-printed prototypes, **cutting 60 g per unit** while meeting stiffness requirements.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Enhanced chassis thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.
- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Reverse-engineered & locally manufactured an RF waveguide connector, reducing cost by 8× compared to imports.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

Gearbox design — Dec 2025 Personal project
Designed and built a 1:64 multi-stage spur gear gearbox system to power a conveyor platform for scaled vehicle testing.
6-DOF Mimicking Robot— July 2025- Nov 2025 Personal project
Engineered a 6-DOF robotic arm with real-time mirroring, trajectory recording, and repeatable playback of arm movements.
SAE Bicycle Design Competition — 2020 - 2021 SAE INDIA
Developed a planetary gear-based bicycle drivetrain achieving 15% higher torque than a typical bicycle.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, **GD&T, Tolerance stackup**, jigs & fixtures, pneumatic fixtures, 3D Printing
Simulation: Ansys Mechanical & Fluent, Finite Element Analysis, SolidWorks Simulation & Motion Analysis, thermal and vibration analysis
Programming & Embedded systems: Python, MATLAB Simulink & Simscape, C, C++, Arduino
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Standards & Testing: IEC 60068-2-6/-64/-31, FAI, ECR/ECN